

CacheWise 的核心 predictor 没有稳定迁移到我们的 agent traces

论文复现 stability • causal progress • synthetic concurrency • task

一句话结论

metadata 有信号，
但收益被少数极端任务支配；
离可部署 policy 还差
task robustness 与真实
KV pressure。

+0.587 s

SWE • C100 更差

95% CI [+0.120, +1.188]

-72.299 s

TB • aggregate 更好

但 single-task fragile

+4.012 s

Timeout • delete-one 翻转

mixed-integer-programming

16 slides • 15–20 min • development-only evidence

Coding agent 把一次用户任务变成 “LLM $\leftrightarrow$ tool” 的长闭环



36 min

CATraces median session

tail > 2.6 h

20×

tool-triggered turns

median vs user-triggered

21×

prefill / decode ratio

vs chatbot workloads

真正的 eviction 问题不是“多久”，而是“谁最晚回来”

memory pressure ^{now} at time t



Belady-like target

$$j^* = \arg \max \tau_j(t)$$

CacheWise 的关键简化 不必精确预测 duration ;
只要把“最晚回来”的
session 排到第一。

CacheWise 用两个互补机制减少 KV thrashing

① PREFIX-AWARE SCHEDULING

先服务需要新增 KV block 最少的 request

$$a_i(t) = d_i -$$

$k_i(t)$

A 

+2 blocks

B



+7 blocks

C



+4 blocks

结果：少动别人的 prefix，也近似 shortest-job-first

② PREDICTIVE EVICTION

真的要 evict 时，优先释放预计最晚复用的 KV

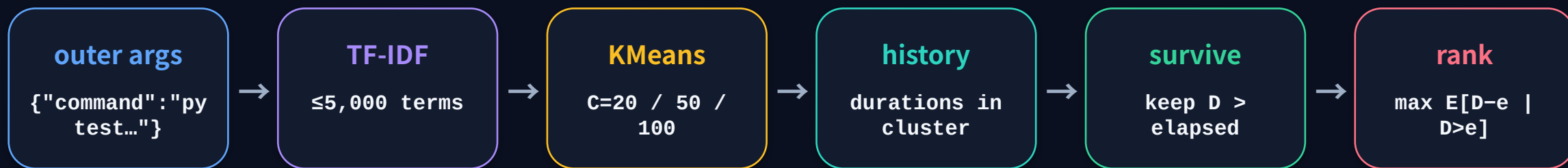
$tool_name + tool_args + elapsed$



$E[remaining \mid survived \text{ to } elapsed]$

vLLM prototype $\approx 2,500$ Python LoC · 每 3 engine iterations 重建 eviction heap

Predictor 是 survival estimator , 不是普通 duration regression



小例子

cluster history: 2s, 5s, 40s, 90s

`elapsed = 10s → survivors = 40, 90`

prediction = mean(30, 80) = 55s remaining

BOUNDARY

论文的 **tool_args = 完整外层 payload**

不解析 shell AST · 不拆 pipeline · 不观察
execve child · 不使用 per-process counters

论文的 $3.5\times$ headline $\neq$ argument predictor 的 isolated gain

PAPER • FULL SYSTEM

2× H200 • Qwen2.5-Coder-32B • TP=2

- CATraces random 80/20 sessions
- 30 / 40 / 50 concurrent sessions
- prefix-aware scheduling + predictive eviction

2.7–3.5×
session time

2–2.6×
fewer
evictions

≤19%
C100 isolated

OURS • PREDICTOR TRANSFER

只复现一个必要子命题

**“whole args 是否比 tool-name
更会排 victim？”**

- 同样的 global / tool / C20 / C50 / C100
- 同样的 conditional remaining 与 C100 primary
- 没有实际 KV capacity / eviction / JCT

**如果 predictor gate 不过，就没有理由花成本
做 live serving。**

我们的 traces 保留 raw args , 但物理条件与论文不同

corpus	fit	evaluation	tool gaps	observed ranking
SWE-ReBench	100 sessions	277 sessions	4,175 → 12,771	2,410 events • max c=2
Terminal-Bench	139 declared 136 w/ gaps	100 declared 99 w/ gaps	1,888 → 1,227	synthetic only

WE HAVE

raw outer tool args
≈0.5s causal CPU/network
samples

WE DO NOT HAVE

recorded shared KV pressure
actual eviction / recompute
feedback

SYNTHETIC

只平移真实 trace 的到达时间
内部 gap duration 保持不变

唯一主指标：选错 victim 会损失多少“可释放时间”

ranking event @ t

A	actual remaining	40s	tool predicts 10s
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B	actual remaining	15s	tool predicts 20s
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C	actual remaining	4s	tool predicts 5s
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tool chooses B → **regret = 40 - 15 = 25s**

定义

$$\text{regret} = \max(\text{actual remaining}) - \text{chosen actual remaining}$$

0 = oracle choice

$$\Delta = \text{mean regret}(\text{candidate}) - \text{mean regret}(\text{tool})$$

GO only if 95% CI upper < 0

口径：hypothetical victim ordering
≠ eviction count ≠ KV bytes ≠ JCT

五个冻结实验逐步排除四种解释

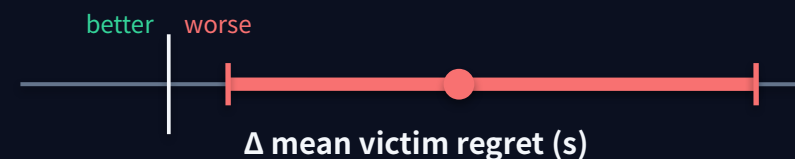
1	SWE observed overlap	whole args?	+0.587s	NO-GO
2	causal progress	live counters?	+0.505s	NO-GO
3	SWE synthetic c32	overlap too sparse?	+1.847s	NO-GO
4	TB synthetic c32	corpus-specific?	-72.299s	FRAGILE
5	exact timeout + LOTO	simple signal stable?	+4.012s*	NO-GO

* step 5 reports the predeclared worst delete-one-task result; aggregate timeout $\Delta = -17.271s$.

SWE : argument clustering 增大 regret , 主 gate 失败



PRIMARY C100 – tool

+0.587 s95% CI
[+0.120, +1.188]

rare tail dominates

top-1: 93.98% →
94.07%p99 regret: 17.3s →
35.5s

90 choice changes: 46 help / 44 harm • harm 1,803s vs help 390s

SWE 失败不是“cluster 太小”，而是相似命令跨 repo 的尺度不同

OBSERVED

syntactically similar pytest / test-suite cluster

FIT cluster max **3–29s**

HELD-OUT still active **108–300s**

C100 predicts 0–5s remaining **tool-name predicts 20–53s**

median harmful-cluster surviving support = 36
→ 不是简单 singleton 问题

SUPPORTED INFERENCE

TF-IDF 识别“在跑测试”，
却看不到 suite
scale、dependency/cache
state、repo-specific work。

NOT TESTED

环境状态特征是否能泛化？

repo size • test inventory •
dependency/cache state • process phase
不能在已暴露 fresh-277 上继续调 tokenization / support
threshold。

Progress counters 改善 MAE，却把 ranking 决策推向错误方向



point prediction

57.43→55.24

MAE improves

+26.25→-16.83

signed error flips

decision boundary

8 / 63

help / harm changes

32.9 / 1,249

help / harm seconds

OBSERVED: 71 changes 都从 still-running gap 转向新到达 gap。

INFERENCE: model 对活跃长任务变得过度乐观；
fit metric 与 policy utility 方向相反。

把 concurrency 合成到 c32，SWE 仍然失败



arrival shifts only; each trace keeps its internal gaps + durations

32

arrival seeds

22.2

mean live sessions

36–42

max live sessions

SWE synthetic-c32

tool **57.961s** C100 **59.808s**

$\Delta +1.847s$ 95% CI [+1.087, +2.646]

24 harmful / 8 helpful schedules

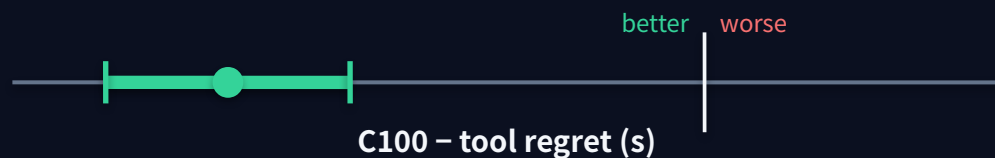
已排除：natural SWE overlap 太稀疏

未排除：真实 contention 会改变 duration / arrival / eviction feedback。

TB aggregate 反转为正，但几乎由 hdfs-deployment 扛住



↓ lower is better



29 / 32 schedules improve

POST-HOC SENSITIVITY

remove hdfs-
deployment

+26.722 s

95% CI [+17.355, +37.789] • harmful 28/32

OBSERVED

hdfs-deployment **2 × ≈ 600s**
download exec

C100 从 command text + explicit timeout=600
识别长 survivor。

C50 regret 113.224s < C100 122.334s ; 但 C50 不是
primary，不能事后改选。

Timeout 只是 ceiling：task-stability gate 失败

AGGREGATE PASSES

tool 194.633s → timeout 177.362s

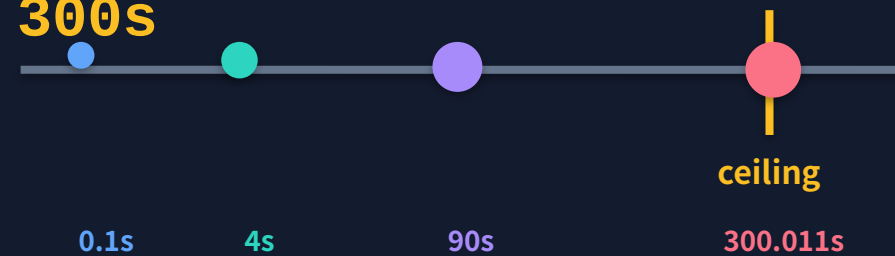
Δ -17.271s • CI [-29.739, -4.296] • 20/32 helpful

TASK STABILITY FAILS

delete mixed-integer-programming **+4.012s**

仅 13/32 schedules helpful • 99 个 task 全穷举，只有这一个翻转符号

timeout =
300s



同一个 timeout 值覆盖多个数量级的实际 work。

OBSERVED

600s signature 提供最大 aggregate benefit；
300s signature 若没有这个跑满 300s 的 solver call，则净有害。

结论：停止 tuning；下一次证据必须跨过 task 与真实压力两道门

① insight survives

相对 next-reuse ordering 是正确
systems objective；tool metadata
确实含 tail signal。

② estimator fails

whole-args、progress、synthetic
SWE、exact timeout 都未通过稳定
gate。

③ scope matters

offline regret 不是 eviction/JCT；
TB aggregate pass 也不能覆盖
single-task fragility。

NEXT EVIDENCE RUNG

freeze current NO-GO

no post-hoc clause/text rescue



new independent tasks

task-grouped gate;
environment-scale hypothesis



task-stable ranker

charge inference + scheduling
overhead



live KV pressure

30–50 sessions; actual
evictions + JCT

最终 takeaway

“use tool args” 不是方法贡献；可发表的问题是：什么 inference-time state 能跨 task 稳定
改善一个真实、计费完整的 KV decision？